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Bitcoin Market Sentiment and Trader Performance Analysis

Objective

The objective of this analysis is to investigate the relationship between Bitcoin market sentiment and trader performance using historical trading data from Hyperliquid and the Bitcoin Fear & Greed Index. The study aims to determine whether market sentiment influences profitability, trading behavior, and overall trading outcomes. By identifying patterns between sentiment and trading performance, the analysis seeks to uncover insights that can support more informed trading decisions and risk management strategies.

Dataset Description

Two datasets were used in this project. The first dataset is the Bitcoin Fear & Greed Index, which classifies market sentiment into categories such as Extreme Fear, Fear, Greed, and Extreme Greed. The second dataset contains historical trading records from Hyperliquid, including information such as trader accounts, traded coins, execution prices, trade sizes, trade direction, fees, and realized profit and loss (Closed PnL). Together, these datasets provide a comprehensive view of both market sentiment and trader activity.

Data Cleaning and Preprocessing

The trading dataset was examined for missing values, incorrect data types, and inconsistencies. Timestamp fields were converted into a standard datetime format to enable date-based analysis. Relevant columns such as Closed PnL, Size USD, Execution Price, and Fee were validated and prepared for analysis. A new binary variable indicating whether a trade was profitable was also created to support win rate calculations.

The sentiment dataset was cleaned by converting date fields into a consistent format. Sentiment classifications and sentiment scores were retained for subsequent analysis.

The data was checked to ensure there were no duplicate or invalid records that could affect the results.

Merge Methodology

To combine the two datasets, the trading timestamps were converted into trading dates and matched with the corresponding dates in the Fear & Greed Index dataset. A left join operation was used to attach the daily sentiment classification and sentiment score to each trade. This approach allowed every trade to be analyzed within the context of the prevailing market sentiment on that specific day.

Sentiment vs Profitability

The analysis compared trader profitability across different market sentiment conditions. By grouping trades according to sentiment classifications, average profit and loss values were calculated and compared. This helped identify whether traders performed better during Fear, Extreme Fear, Greed, or Extreme Greed periods.

The results indicate that market sentiment has a measurable influence on trading outcomes. Periods of positive sentiment generally corresponded with increased trading activity and stronger profitability, while periods of fear often displayed more cautious trading behavior. These findings suggest that market psychology can significantly impact trader performance.

Win Rate Analysis

Win rate was calculated as the percentage of trades that generated a positive realized profit. By examining win rates across sentiment categories, it was possible to evaluate whether traders were more successful under specific market conditions.

The analysis revealed noticeable differences in win rates between sentiment regimes. Higher win rates were generally observed during favorable market conditions, while periods of fear were associated with lower trading success. This trend suggests that sentiment may influence both market direction and trader decision-making effectiveness.

Analysis

Top Traders

Trader performance was evaluated by aggregating total realized profit and loss for each account. The analysis identified the highest-performing traders and highlighted the concentration of profits among a relatively small group of participants.

The results show that a limited number of traders generated a substantial portion of overall profits. This observation indicates differences in trading skill, risk management practices, and strategy effectiveness across market participants.

Coin Analysis

Profitability was also analyzed at the coin level. Total realized profit and loss was calculated for each traded asset to determine which coins contributed most significantly to trader performance.

The analysis revealed that certain coins consistently generated higher profits and trading activity than others. This suggests that asset selection plays an important role in overall trading success and that some markets provide more favorable trading opportunities.

Long vs Short Analysis

Trading performance was further examined based on trade direction. Buy-oriented and sell-oriented trades were analyzed separately to determine whether one approach was more effective under different market conditions.

The results indicate that trade direction can have a significant impact on profitability. Market sentiment often influenced the effectiveness of long and short positions, with different sentiment environments favoring different trading strategies.

CONCLUSIONS

This study demonstrates that market sentiment has a meaningful relationship with trader behavior and performance. Differences in profitability, win rates, trading volume, and trade direction were observed across various sentiment conditions. The findings suggest that traders respond differently to market optimism and pessimism, which ultimately affects their results.

The analysis also highlights the importance of trader selection, asset selection, and strategic positioning. Top-performing traders consistently outperformed the broader trading population, while certain coins and trading approaches contributed disproportionately to overall profitability.

RECOMMENDATIONS

Traders should incorporate market sentiment indicators into their decision-making process to better understand prevailing market conditions. Monitoring sentiment can help identify periods of elevated risk and potential opportunity. Risk management practices should be adjusted according to sentiment regimes, particularly during periods of extreme fear or extreme greed.

Future analysis could be enhanced by incorporating additional market indicators, leverage information, volatility measures, and machine learning techniques to develop predictive models for trader performance. Combining sentiment analysis with quantitative trading signals may provide a more comprehensive framework for improving trading outcomes.